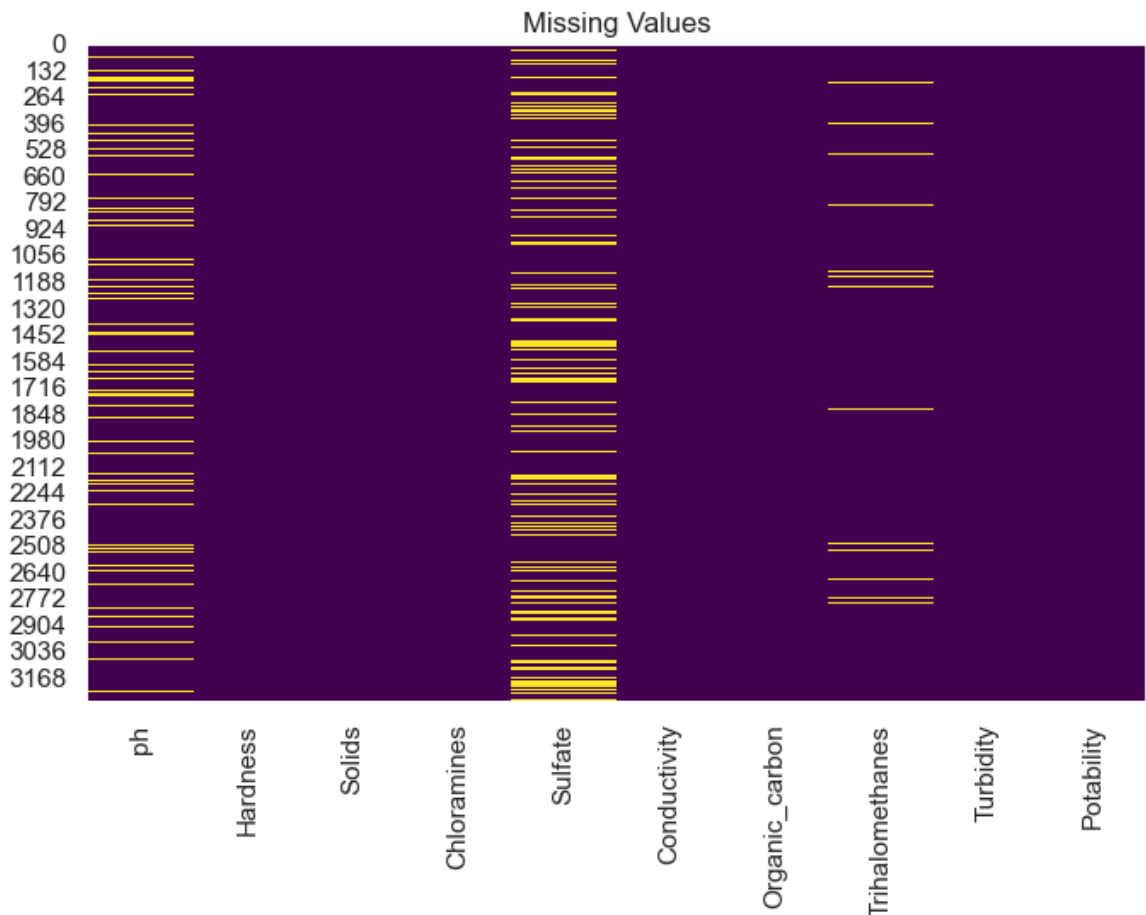


Graph Analysis

1) Missing Value Heatmap

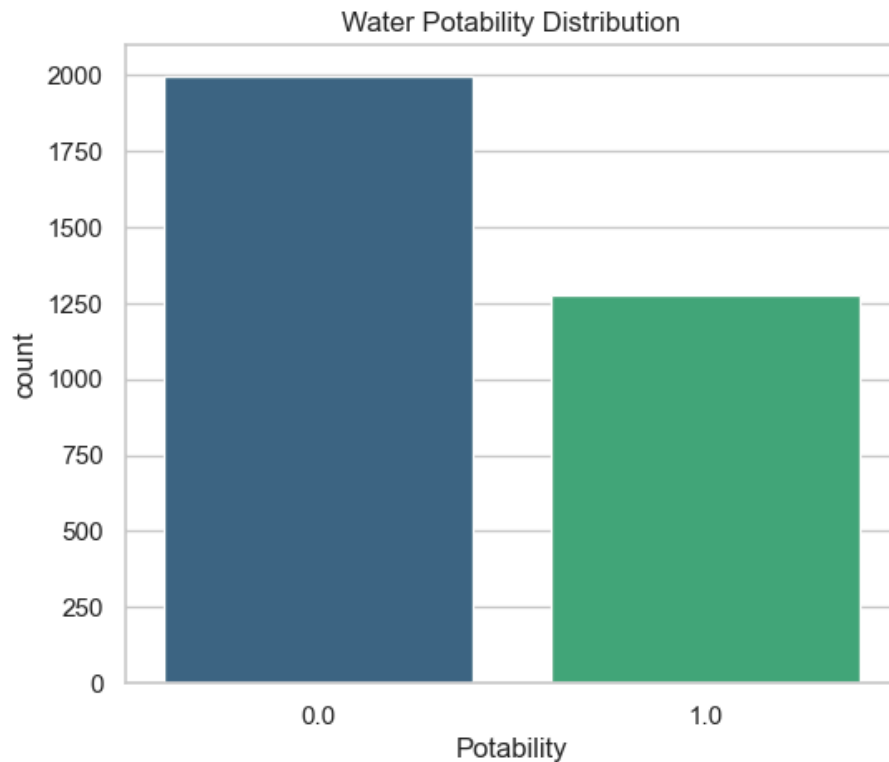


The missing value heatmap provides an overview of incomplete observations in the dataset. It reveals that the pH, Sulfate, and Trihalomethanes features contain missing values, while the remaining features are complete.

Since machine learning algorithms cannot directly process missing values, these values were handled before model training. Median imputation was chosen because it is less sensitive to extreme values and outliers compared to the mean. This approach preserves the overall distribution of the data while ensuring that no valuable records are discarded.

Proper handling of missing values improves the reliability and stability of the predictive models.

2) Class Distribution

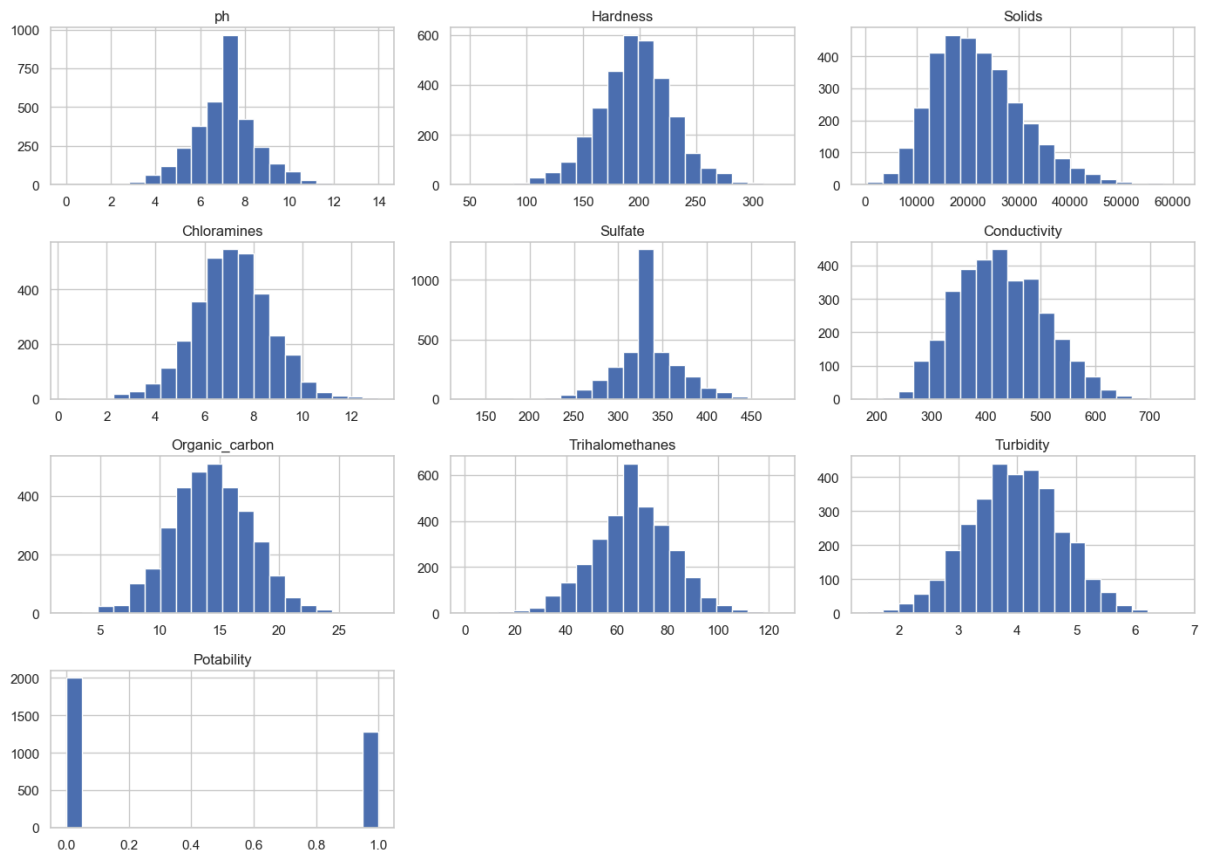


This chart illustrates the distribution of potable and non-potable water samples in the dataset.

The dataset is moderately imbalanced, with a greater number of non-potable (0) samples than potable (1) samples. Training a model on imbalanced data can lead to biased predictions toward the majority class.

To address this issue, the SMOTE (Synthetic Minority Oversampling Technique) algorithm was applied after preprocessing. SMOTE generates synthetic examples of the minority class, resulting in a balanced dataset and improving the model's ability to correctly identify potable water samples.

3) Feature Distributions



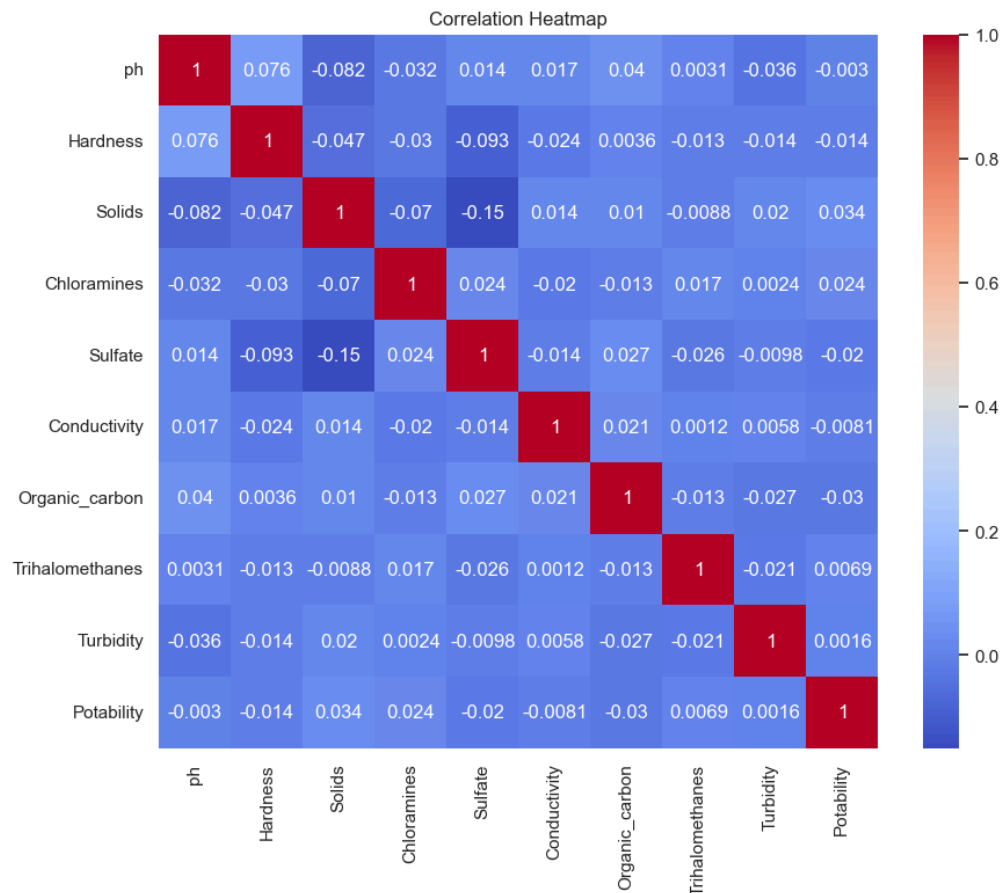
Histograms of Numerical Features

The histograms display the distribution of each numerical feature in the dataset.

Several variables exhibit skewed distributions, indicating that the data is not normally distributed. Features such as **Solids**, **Sulfate**, and **Trihalomethanes** show considerable variability, while others are more concentrated around specific ranges.

Understanding feature distributions helps identify data characteristics and informs preprocessing decisions such as feature scaling. Therefore, **StandardScaler** was applied before training the machine learning models to normalize feature values.

5) Correlation Analysis



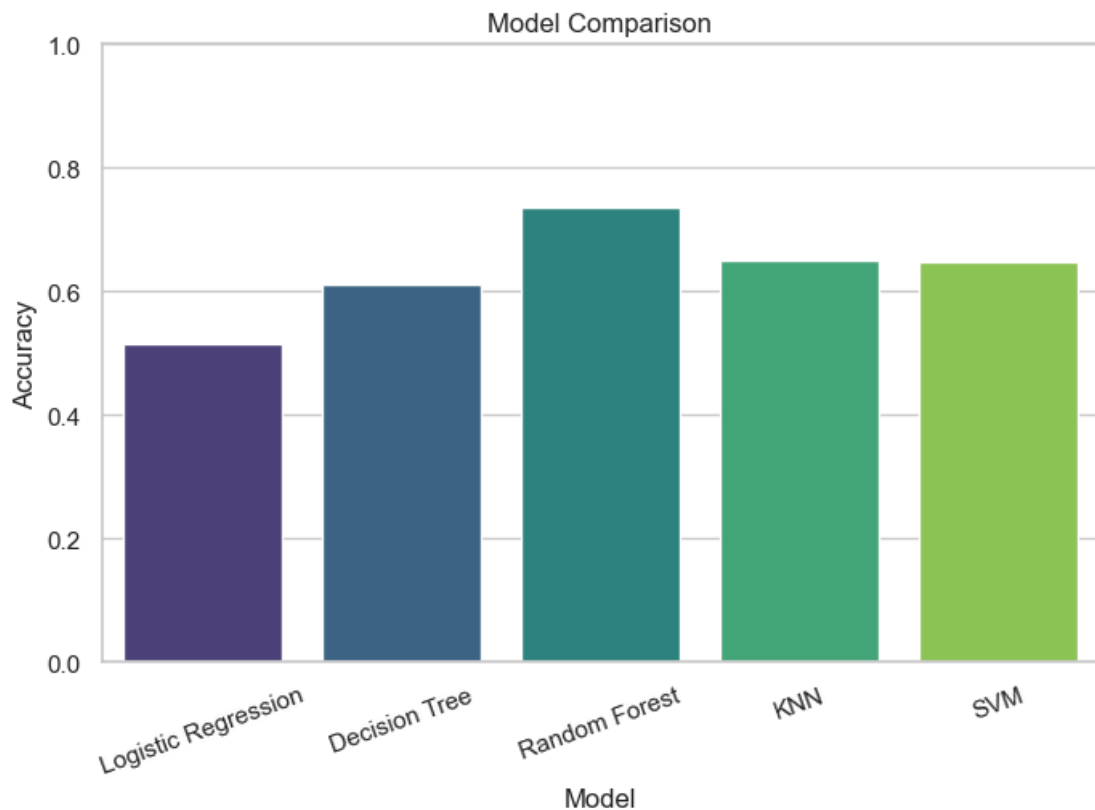
Correlation Analysis

The correlation heatmap illustrates the pairwise relationships among all numerical features.

Most features exhibit weak to moderate correlations, indicating that no single variable strongly predicts another. The absence of severe multicollinearity is beneficial because highly correlated features can negatively affect some machine learning algorithms.

These observations suggest that the prediction task depends on the combined influence of multiple water quality parameters rather than a single dominant feature.

6) Model Performance Comparison



Model Comparison

Five supervised machine learning algorithms were trained and evaluated:

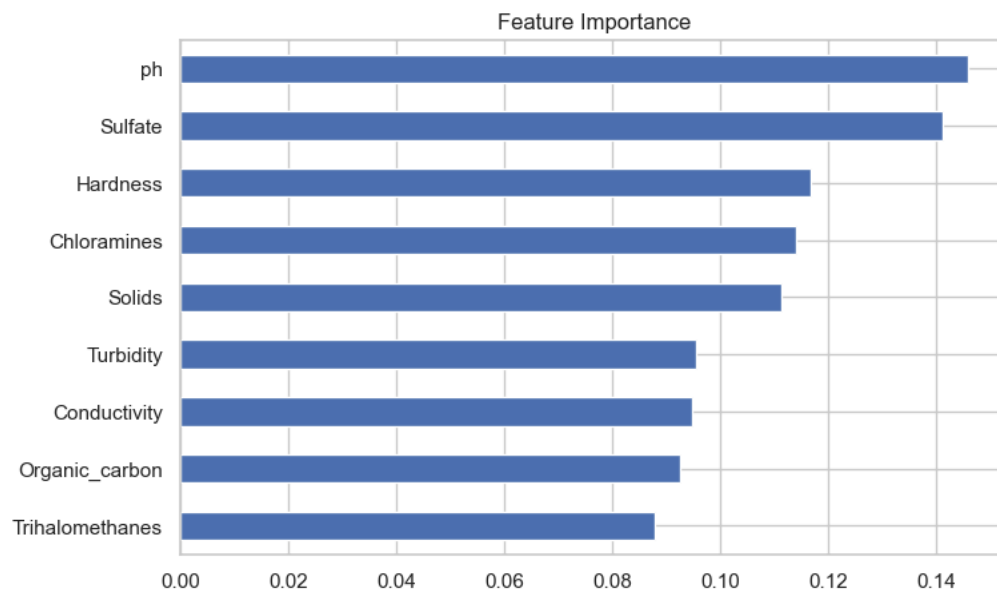
- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Among these models, Random Forest achieved the highest performance with an accuracy of 73.63%, making it the most effective classifier for this dataset.

KNN and SVM produced competitive results, while Logistic Regression achieved the lowest accuracy because the relationship between water quality parameters and potability is highly nonlinear.

This comparison demonstrates the importance of evaluating multiple algorithms before selecting the best-performing model.

7) Feature Importance



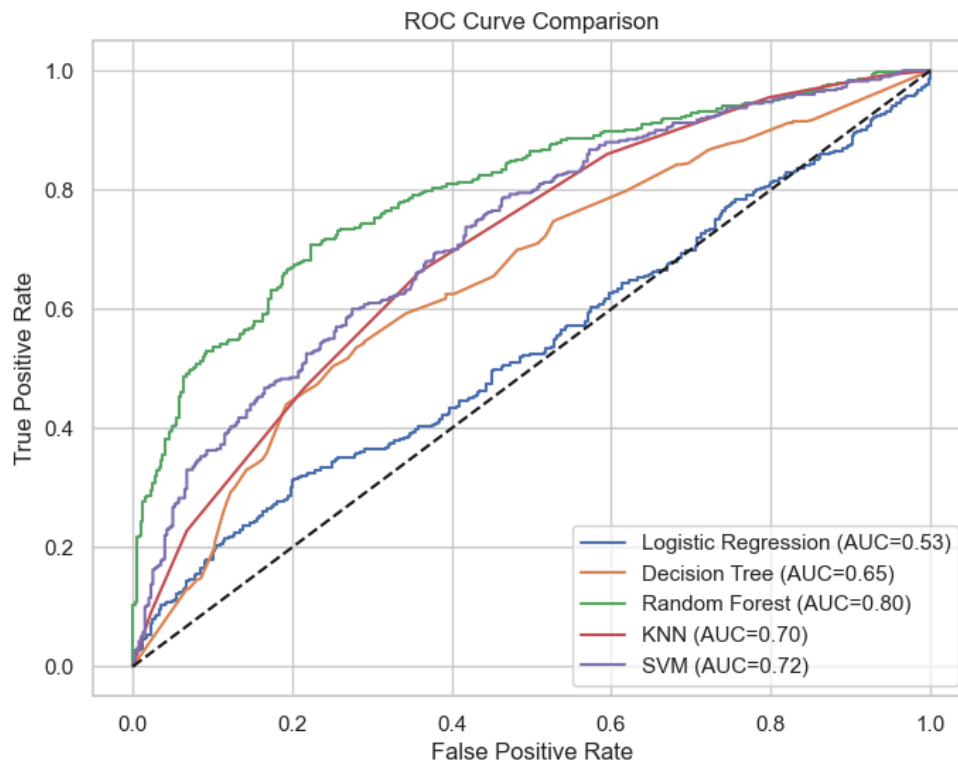
Important Features

Feature importance analysis was performed using the trained Random Forest model.

The graph ranks water quality parameters according to their contribution to the prediction process. Features with higher importance scores have a greater influence on determining whether a water sample is potable.

Understanding feature importance improves model interpretability and provides insight into which physicochemical properties are most influential in water quality assessment.

8) ROC Curve Comparison



ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve evaluates the classification performance of each model across different decision thresholds.

A model with a larger Area Under the Curve (AUC) demonstrates better discrimination between potable and non-potable water samples.

The ROC comparison confirms that Random Forest consistently provides the strongest predictive performance among the evaluated algorithms, achieving a better balance between true positive and false positive rates.